

Section A: Pure Mathematics [40 marks]

- 1 (a) (i) Without using a calculator, solve the simultaneous equations
- $$\begin{aligned} 3y - 2x &= 5 \\ 8x^2 + y &= 4 \end{aligned} \quad [3]$$
- (ii) Hence or otherwise, solve the inequality
- $$12 - 24x^2 \geq 2x + 5 \quad [2]$$
- (iii) Hence solve the inequality
- $$12 - 24(2^{2x}) \geq 2^{x+1} + 5 \quad [2]$$
- (b) Find the range of values of a such that the expression $x^2 - 2ax + 3a + 4$ is always positive. [3]

- 2 The functions f and g are defined by

$$f : x \mapsto \ln(x^2 + 1), x \in]$$

$$g : x \mapsto x^2, x \in], x < 0$$

- (i) Show that g^{-1} exists and find g^{-1} . [4]
- (ii) Show that gf does not exist. [2]

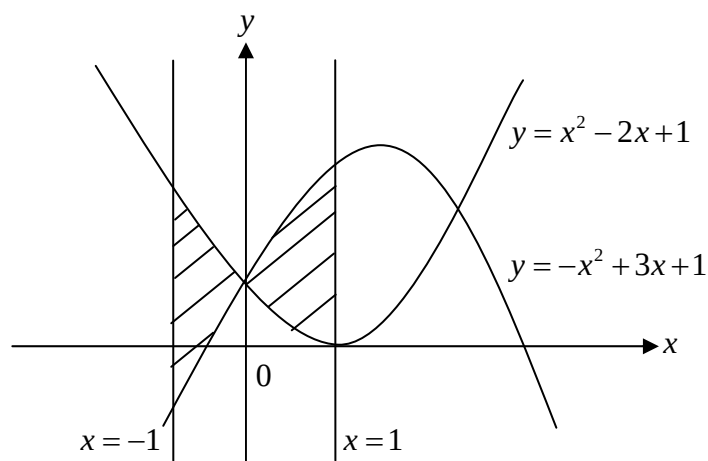
Express each of the following functions in terms of f , g and g^{-1} as appropriate.

- (iii) $x \mapsto \ln(x^4 + 1), x \in], x < 0$ [1]
- (iv) $x \mapsto x, x \in], x < 0$ [1]
- (v) $x \mapsto x, x \in], x > 0$ [1]

- 3 Integrate the following with respect to x

- (i) e^{3x+2} [1]
- (ii) $\frac{(x+1)^2}{\sqrt{x}}$ [2]

4



Find the area bounded by the curves $y = -x^2 + 3x + 1$ and $y = x^2 - 2x + 1$ and the lines $x = -1$ and $x = 1$. [3]

5

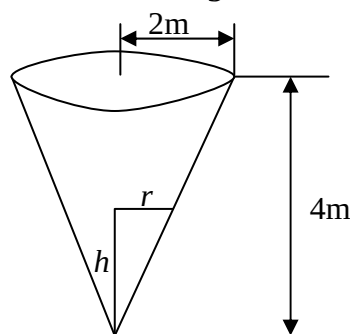
(a) Differentiate $\ln\left(\frac{e^{2x}}{\sqrt{e^{2x}+1}}\right)$ with respect to x . [3]

(b) The equation of a curve, C , is given by $y = 3 + 2x - x^2$. Find the equation of the normal at the point where $x = 2$. Given that the normal cuts C at another point, E , find the coordinates of point E . [5]

(c) An inverted cone was initially filled with water and a hole was made at the bottom of the cone. The cone has a vertical height of 4 m and radius 2 m. The volume of the water in the cone is decreasing at a rate of $\frac{\pi}{12} \text{ m}^3/\text{s}$.

(i) Show that the volume of the water in the cone can be expressed as $\frac{1}{12}\pi h^3$. [2]

(ii) Hence, find the rate at which the depth of the water is decreasing after 37 seconds. (Volume of cone = $\frac{1}{3}\pi r^2 h$) [5]



Section B: Statistics [60 marks]

- 6** In an office, on average, 2 staff in 5 drive a car.
- (i) Calculate the probability that in a randomly chosen sample of 15 staff, more than 8 staff drive a car. [2]
- (ii) A random sample of 150 staff is selected. Using a suitable approximation, find the probability that the number of staff who do not drive a car is not more than 100 but greater than 80. [3]
- 7** The mass of a randomly chosen orange in a plantation, X g, is normally distributed with mean of 500 g and standard deviation of 16 g. A random sample of size n is taken and weighed. If $P(489 < \bar{X} < 505) > 0.999$, find the minimum value of n . [3]
- 8** (a) State briefly a reason for taking samples. [1]
- (b) A military camp consists of 100 soldiers of whom 10 are lieutenants, 20 are majors, 30 are sergeants and the rest are privates. A sample of 20 soldiers is required to form a committee that looks into the welfare of the soldiers in the camp.
- Explain how this may be obtained using
- (i) Systematic sampling [2]
- (ii) Stratified sampling [2]
- (iii) State, with a reason, which of the above 2 methods is more suitable in the context of the question. [1]
- 9** (a) In a finite sample space, the random events, A , B and C are related as follows:
The events A and B are mutually exclusive and events A and C are independent.
If $P(A) = \frac{1}{2}$, $P(B) = \frac{5}{12}$, $P(A \cup C) = \frac{2}{3}$, $P(B \cup C) = \frac{2}{3}$,
Find, giving your answers in fractions,
- (i) $P(A \cup B)$ [2]
- (ii) $P(C)$ [2]
- (iii) $P(B \cap C)$ [2]
- and state whether B and C are independent events. [1]

- 9 (b) A bag contains 4 black balls and 8 white balls. Three balls are drawn at random, one by one and without replacement.
- (i) Construct the probability tree diagram illustrating the information given. [2]
- Find, exactly,
- (ii) the probability of drawing 3 black balls. [1]
- (iii) the probability of drawing at least 2 white balls. [2]
- (iv) the probability that the first ball drawn is black given that at least 1 white ball is drawn. [3]
- 10 At Wimbledon college, the marks obtained in the Mathematics examination denoted by X may be assumed to be normally distributed with mean μ and standard deviation σ .
- (i) Given that $P(X > 62) = 0.0228$ and $P(X < 55) = 0.0668$, find the value of μ and σ , giving your answers correct to the nearest integer. [5]
- (ii) Find the probability such that the total marks obtained by 3 randomly chosen students exceed 180. [2]
- At Manchester college, the marks obtained in the same examination is denoted by Y . Y may be assumed to be normally distributed with mean 62 and standard deviation 3.
- (iii) What is the probability that the total marks in the Mathematics examination scored by two students from Manchester college is more than twice the marks scored by one student from Wimbledon college? [4]
- (iv) State one assumption used. [1]
- 11 The mean mass of a plasma television set is said to be 65 kg. A random sample of 200 sets is taken and their masses are measured. The masses, x kg, are summarised by $\sum x = 12\,000$ and $\sum (x - 60)^2 = 218\,192$.
- (i) Find the unbiased estimates of the population mean and variance [3]
- (ii) Assuming a normal distribution, test at 5% significance level whether there is a change in the population mean mass. [4]
- (iii) Explain what is meant by “5% level of significance” in the context of the question. [2]

- 12 Eleven psi readings were obtained in the month of August 2008 (from 21st to 31st August) as follows in Country A.

psi index (y)	20	33	38	39	44	42	48	46	51	46	52
Time / Days (x)	1	2	3	4	5	6	7	8	9	10	11

The psi index is used as a measure of the haze condition in Country A and the index becomes higher when the haze condition worsens. It was also known that the haze conditions in Country A starts to worsen from August until the end of September every year.

- (i) Sketch the scatter diagram for the set of data. [2]
- (ii) It was known that one of the data points was wrongly recorded. Identify the data point and find the linear moment correlation coefficient for the remaining ten data points. Comment on the correlation coefficient obtained. [3]
- (iii) A meteorologist wanted to predict the date when the psi index will hit a hazardous level of over 118.
 - (a) Using the ten data points, find an appropriate regression line, justifying your choice of the regression line used. [2]
 - (b) Find the date at which the psi index will hit over 118. Comment on this value obtained. [3]

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